TRT 09

Gonadorelin

Synthetic copy of GnRH (LHRH), the 10-amino-acid peptide hormone made by the hypothalamus



CLASS
Peptide hormone

STUDIED FOR
Hypogonadism, sperm production

HUMAN EVIDENCE
FDA approved (1982); small studies in men

US STATUS
FDA approved 1982; human products discontinued

STUDIED FOR

- **Sperm production** — HUMAN
- **Testosterone, LH, FSH** — HUMAN
- **Testis volume** — HUMAN
- **Bone density** — HUMAN

No studies with TRT. No published human study of gonadorelin given with testosterone therapy was found. No US product is labelled for that use; it is off-label.

HOW IT WORKS IN THE BODY

Gonadorelin ▶ Pituitary ▶ LH and FSH ▶ Testes

- **Same sequence** as the body's own GnRH
- **Induces release** of endogenous LH from the pituitary (label)
- **LH and FSH** drive testosterone and sperm production in the testes
- **Given in pulses** by pump in men who lack their own GnRH

EVIDENCE & WHAT'S KNOWN

EVIDENCE

FDA APPROVED
LEVEL 5 OF 5

- Approved as Factrel (1982), Lutrepulse (1989)
- Pump studies in 22 to 28 men, not randomized
- No human studies alongside TRT

NOT YET ESTABLISHED

- Use with testosterone therapy
- Fertility restoration after TRT
- Long-term safety in men

QUICK PRACTICAL INFO

HALF-LIFE
Not established

ROUTES STUDIED
Under the skin by pulsed pump in studies; injection, nasal (EU)

PHARMACOKINETICS
Not established

SOLD AS
No US human product; compounding (503B) under FDA review

MONITORED IN TRIALS
Sperm, testis volume, hormones, bone density

SPORT (WADA)
Prohibited, class S2 (males)

REGULATORY

FDA	Approved 1982; human products discontinued
FDA	503B category 1, under evaluation
EMA	Authorised in 5 EU member states
WADA	Prohibited at all times in males (S2)

REFERENCES

1. FDA, Drugs@FDA record for Factrel (NDA 018123), accessed Nov 2026
2. Am J Mens Health, gonadorelin pump vs hCG/hMG in hypogonadism, Jan 2019
3. WADA, 2026 Prohibited List

Adonis Men's Health · checked Nov 2026. Research and regulator records only. Not medical advice.

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